

CSC 211: Computer Programming

Structs

Dr. Michael Conti

Department of Computer Science and Statistics
University of Rhode Island

Summer 2026



Original design and development by Dr. Marco Alvarez

Structures

- Definition is generally outside any function
 - new 'data type' will be available to all code that follows
- Structures can be declared in the same way as basic data types
- Can also use { } notation for initialization
- Use the **dot operator** for accessing data members

2

Structures

```
struct structureName {  
    member1;  
    member2;  
    member3;  
    .  
    .  
    .  
    memberN;  
};
```

Structures in C++ are user defined data types which are used to store multiple items (members) of possibly different data types

3

Example

```
// defining the struct  
struct Point {  
    int x;  
    int y;  
};  
  
int main() {  
    // creating a variable  
    struct Point p1;  
}
```

4

Initializing ...

```
// defining the struct
struct Point {
    int x;
    int y;
};

int main() {
    // initializing (follows order)
    struct Point p1 = { 10, 20 };
}
```

5

The dot operator

```
#include <iostream>

struct Point {
    int x;
    int y;
};

int main() {
    struct Point p1 = { 10, 20 };
    p1.x += 5;
    std::cout << p1.x << ' ' << p1.y << '\n';
}
```

6

The dot operator

```
#include <iostream>

struct Point {
    int x;
    int y;
};

int main() {
    struct Point p1 = { 10, 20 };
    struct Point p2 = { 30, 40 };
    struct Point p3 = { 50, 60 };
    p1.x += 5; p2.y += 10; p3.y += 15;
}
```

7

DISPLAY 10.2 Member Values

```
1 struct CDAccount
2 {
3     double balance;
4     double interestRate;
5     int term; //months until maturity
6 };
7 int main( )
8 {
9     CDAccount account;
10     ...
11
12
13     account.balance = 1000.00;
14
15
16     account.interestRate = 4.7;
17
18
19     account.term = 11;
20
21
22
```

State	balance	interestRate	term
Initial	?	?	?
After account.balance = 1000.00;	1000.00	?	?
After account.interestRate = 4.7;	1000.00	4.7	?
After account.term = 11;	1000.00	4.7	11

from: Problem Solving with C++, 10th Edition, Walter Savitch

8

Array of structures

```
#include <iostream>

struct Point2D {
    double x;
    double y;
};

int main() {

    Point2D mypoint;
    Point2D myarray[5];

    mypoint.x = 10;
    mypoint.y = 20;

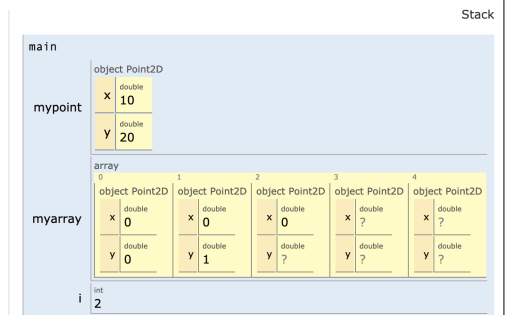
    for (int i = 0 ; i < 5 ; i++) {
        myarray[i].x = 0;
        myarray[i].y = i;
    }
}
```

9

pythontutor.com

C++ (gcc 4.8, C++11)
EXPERIMENTAL! known limitations

```
1 struct Point2D {
2     double x;
3     double y;
4 };
5
6 int main() {
7     struct Point2D mypoint;
8     struct Point2D myarray[5];
9
10    mypoint.x = 10;
11    mypoint.y = 20;
12
13    for (int i = 0 ; i < 5 ; i++) {
14        myarray[i].x = 0;
15        myarray[i].y = i;
16    }
17
18 }
```



10

Arrays and Structures

- When using arrays as structs member, the index goes at the end
 - student.grades[i]
- When using structs as arrays elements, the index goes after the struct name
 - students[i].finalGrade

11

Functions

```
// defining the struct
struct Point {
    int x;
    int y;
};

void distance(Point P1, Point P2);
```

12

Passing structures to functions

- A struct can be passed as a parameter either by value or by reference

```
void printPoint(Point &somePoint){  
    std::cout << somePoint.x;  
    std::cout << somePoint.y;  
};
```

- A function can return a value of type struct

```
Point incrementPoint(Point somePoint){  
    somePoint.x+=1;  
    somePoint.y+=1;  
    return somePoint;  
};
```

13

Passing structures to functions

DISPLAY 10.1 A Structure Definition ..

```
1 //Program to demonstrate the CDAccount structure type.  
2 #include <iostream>  
3 using namespace std;  
4 //Structure for a bank certificate of deposit:  
5 struct CDAccount  
6 {  
7     double balance;  
8     double interestRate;  
9     int term; //months until maturity  
10 };  
11  
12  
13 void getData(CDAccount& theAccount);  
14 //Postcondition: theAccount.balance and theAccount.interestRate  
15 //have been given values that the user entered at the keyboard.  
16  
17  
18 int main( )  
19 {  
20     CDAccount account;  
21     getData(account);  
22  
23     double rateFraction, interest;  
24     rateFraction = account.interestRate / 100.0;  
25     interest = account.balance * rateFraction * (account.term / 12.0);  
26     account.balance = account.balance + interest;  
27  
28     cout.setf(ios::fixed);  
29     cout.setf(ios::showpoint);  
30     cout.precision(2);  
31     cout << "When your CD matures in "  
32     << account.term << " months,\n"  
33     << "it will have a balance of $"  
34     << account.balance << endl;  
35     return 0;  
36 }  
37  
38 //Uses iostream:  
39 void getData(CDAccount& theAccount)  
40 {  
41     cout << "Enter account balance: $";  
42     cin >> theAccount.balance;  
43     cout << "Enter account interest rate: ";  
44     cin >> theAccount.interestRate;  
45     cout << "Enter the number of months until maturity\n"  
46     << "(must be 12 or fewer months): ";  
47     cin >> theAccount.term;  
48 }
```

from: Problem Solving with C++, 10th Edition, Walter Savitch

14

Be careful of same member names

```
// defining the struct  
struct Point {  
    int x;  
    int y;  
};  
  
struct Character {  
    int x;  
    int y;  
    std::string name;  
};
```

Compiler can keep track but it's harder for humans

15

Structs and Pointers

```
struct Books {  
    std::string title;  
    std::string author;  
    std::string subject;  
    int book_id;  
};
```

16

Pointers and Structs

- You can define pointers to structures in very similar way as you define pointer to any other variable

```
struct Books *struct_pointer;  
struct Books Book1;
```

- Now, you can store the address of a structure variable in the above defined pointer variable.

```
struct_pointer = &Book1;
```

17

Structs and Pointers

```
void printBook( struct Books *book ) {  
    std::cout << "Book title : " << (*book).title;  
    std::cout << "Book author : " << (*book).author;  
    std::cout << "Book subject : " << (*book).subject;  
    std::cout << "Book id : " << (*book).book_id;  
}  
  
struct Books {  
    std::string title;  
    std::string author;  
    std::string subject;  
    int book_id;  
};  
  
int main(){  
    struct Books Book1;  
  
    Book1.title = "Learn C++ Programming"  
    Book1.author = "Chand Miyan"  
    Book1.subject = "Computer Science"  
  
    printBook( &Book1 );  
}
```

18

Example

- Write a Student struct that contains
 - ✓ Name
 - ✓ StudentID
 - ✓ Major
- Implement functions:
 - ✓ void buildStudent(Student &someStudent)
 - Initialize member variables of student Struct
 - ✓ void changeMajor(Student &someStudent);
 - Change the major of a student structure
 - ✓ void printStudent(Student &someStudent);
 - Prints out all member variables of student structure

19